

EEG Classification for Motor Imagery Mental Tasks Using Wavelet Signal Denoising

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Abstract – Brain-Computer Interfaces (BCIs) are an approach that enables humans to interact with their surroundings by brain generated control signals. Electroencephalographic (EEG) signals that records electrical activity through the scalp might contain superfluous artifacts suppressing some valuable information. Thus the EEG signal denoising is an important stage of the EEG data analyses. The paper presents an experimental comparison of several classification approaches for 2-class motor imagery EEG data classification and explores the influence of wavelet signal denoising on the classification accuracy.

Keywords – brain computer interface, classification, electroencephalography, 2-class motor imagery, wavelet signal denoising

I. INTRODUCTION

The human cerebrum produces diverse electrical waves over various degrees of awareness of the mind. Electroencephalogram (EEG) signals of the cerebrum can be utilized to screen neural exercises [1]. EEG records electrical movement over regions of the scalp.

The EEG signals can be processed utilizing traditional strategies [2], however, the diversity of the EEG signal could not be possibly mapped by using linear analytical procedure thus techniques based on different frequency transformations are used for analyzing power spectral density and frequency components of the EEG signals.

Suppression in EEG signals could mean possible loss of responsiveness over a period [3]. Since the wavelet entropy is an important variable to consider while investigating the degree of awareness of the human cerebrum wavelet decomposition is widely used in different applications of EEG signals analyzes as it efficiently represents EEG signals by decomposing each signal into frequency sub-bands [4].

Brain computer interface (BCI) is a system that enables humans to interact with their surroundings via employing control signals generated from electroencephalographic (EEG) activity without intervention of peripheral nerves and muscles.

The paper presents an experimental comparison of several classification approaches for 2-class motor imagery EEG data and explores the influence of wavelet based signal denoising on the classification accuracy adopted at the feature extraction stage.

II. RELATED WORK

EEG is the recording of electrical activity through the scalp over a short period of time through multiple electrodes and is considered as the most common method for brain signals recording because it has high temporal resolution, it is easy to use, safe and affordable [5]. The EEG signals might contain superfluous artifacts that may hide some

valuable information in the signal thus making EEG signal denoising important stage of the EEG data processing and analyses.

Wavelets have demonstrated to be a successful instrument for signal analyses and the field of wavelet analysis continues to be heavily researched and has many applications to problems in signal processing. Wavelet analysis has very useful properties such as unconditional basis, i.e. for many signals the expansion coefficients will decrease very quickly. It also generates a more accurate local description of the signal and is more flexible and easy to compute than alternatives such as the Gabor transform [6].

Utilization of discrete wavelet transform for EEG signal denoising is widely studied research area for both medical analyses as well in BCI implementations. In [7] a novel feature extraction algorithm is proposed for extracting event related potentials from the EEG signals using wavelet sub band entropy used in BCI applications. An approach for analyzing EEG signal for anesthesia levels identification based on wavelet entropy is presented in [8]. In [9] feature extraction (denoising) of evoked potentials responses is used by wavelet coefficients scaling based on the phase variance across successive sub averages. An improved empirical approach for EEG signal artifacts is described in [10] that converts single channel EEG into multichannel using Ensemble Empirical Mode Decomposition (EEMD) filtering based on Independent Component Analysis (ICA) and Double Density Wavelet Transform. ICA and different wavelets (Symmlet, Coiflet, Haar, Daubechies, bi-symmetrical) are evaluated for EEG signal denoising [11, 12, 13]. An approach for removing artifacts from multichannel EEG data by a combination of ICA and wavelet-based noise reduction is described in [14]. A non-linear denoising by different discrete wavelet functions using thresholding technique are evaluated in [15]. Morphological feature extraction is demonstrated to substantially reduce the noise in real ECG data [16]. ECG denoising by low-distortion adaptive Savitzky-Golay filtering based on discrete curvature estimation is suggested in [17]. A feature selection approach based on relative energy extracted by wavelet transformation is described in [18]. A feature extraction approach utilizing construction of noisy EEG signal by raw EEG signal normalization and linear mixing with the normalized EOG signal and later noisy EEG signal decomposition by using different wavelets is proposed in [19]. Real-time wavelet-based single-channel EEG signal denoising implemented on Xilinx FPGA is described in [20]. In [21] statistical approach for removing artifacts from EEG recordings is proposed through wavelet transform without considering SNR calculation. Automatic feature selection using dual tree complex wavelet transform is used for classification of imagery left and right hand movements from the EEG signals [22].

The goal of the paper is to evaluate the influence of wavelet based denoising when implemented at the feature selection stage of the EEG data processing pipeline for 2-class motor imagery EEG data.

III. EEG DATA PROCESSING PIPELINE

The general BCI data processing pipeline utilized for experimental evaluation of several classification approaches based on 2-class motor imagery EEG data is illustrated in Fig. 1. The approach employed follows the data processing pipeline based on the Common Spatial Patterns (CSP) algorithm [23] and consolidates wavelet signal denoising at the feature selection stage.

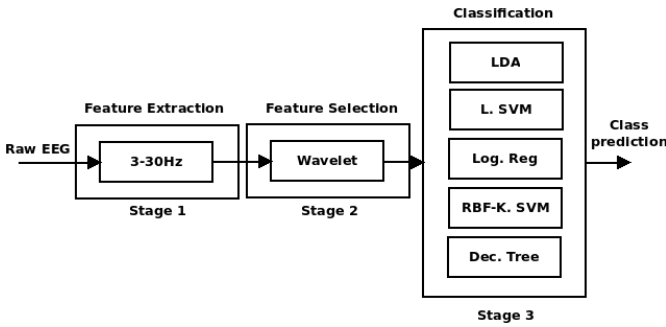


Fig. 1. EEG data processing pipeline.

The raw EEG signal is first frequency filtered using a recurrence separating utilizing a zero-stage Chebyshev Type II channel that covers the recurrence extend in the range of 3 and 30 Hz and providing a recurrence separated matrix. Spatial filtering is also applied by Common Spatial Patterns (CSP) algorithm [23] utilizing spatial filters which maximizes the variances of the filtered time series and renders the classes optimally separable. The frequency-filtered raw data is first projected by the constructed spatial filters and the log of the power of the projected signal is then combined with a feature vector and a bias in order for the prediction to be evaluated [23].

A discrete wavelet transform (DWT) is used at the feature selection stage of the EEG signal processing pipeline for noise reduction and proper feature selection. Signal denoising by DWT is based on successive application of DWT finite number of times and thresholding of the wavelet coefficients, an approach also known as wavelet shrinkage [24]. The wavelet shrinkage involves comparing the coefficients to an arbitrary threshold value and reducing the coefficients value if it is within the absolute value of the threshold: in hard wavelet thresholding the coefficients within the threshold are reduced to zero, effectively cutting them from the signal while soft thresholding subtracts the threshold value from the smaller coefficients rather than setting them to zero. Once wavelet thresholding is used to reduce noise to a satisfactory level the signal may be reconstructed via the inverse discrete wavelet transform (IDWT).

The problem of 2-class motor imagery EEG data classification is a binary classification task thus at the classification stage the classifier is expected to output a real value, whose sign is interpreted as the predicted class.

Five classifiers are experimentally analyzed that were successfully applied for motor imagery classification: Linear Discriminant Analysis (LDA), Linear and RBF-kernel Support Vector Machine (SVM), Logistic Regression and Decision Tree.

IV. EXPERIMENTAL RESULTS

The experimental study of the wavelet based signal denoising of EEG data for 2-class motor imagery classification is based on BCI competition III dataset [24]. The dataset is aimed at solving the problem of motor imagery classification in an asynchronous protocol design, i.e. no cues are provided along with the data to indicate that the subject has switched to a different mental target class. The dataset contains periods in which the user has no control intention and thus the classification algorithm is therefore expected to discriminate between left and right motor imagery as well as the resting state. EEG data are recorded from one subject sitting in a comfortable chair with arm rests and consists of 7 sessions recorded using 118 channels and 120 trials s of motor imagery. Signals are band-pass filtered in the range of 0.05 and 200 Hz and then down sampled at 100Hz before being publicly available for analyses. The mental tasks are as follows: 7 initial sessions; 3 motor imageries (L) left hand, (F) right foot, (Z) tongue; 4 to 7 are given without any cue information (neither target class nor timing) as test set.

A Matlab 2020a implementation of wavelet signal denoising is utilized for the experimental study based on the wavelet Symlets 4 with a posterior median threshold rule. The Symlets 4 wavelet has the properties to be near symmetric, orthogonal and biorthogonal. The decomposition level is set to $\log_2 N$ with N equal to the number of data samples. In the case of the BCI competition III dataset that is used for the experimental study, the level of decomposition is set to 4. An empirical Bayesian method with a Cauchy prior is used for the thresholding value selection. A threshold rule used is based on assumption that the measurements have independent prior distributions given by a mixture model. An example of the raw EEG signal of the dataset is shown in Fig.2 and Fig 3 shows the same signal after wavelet denoising.

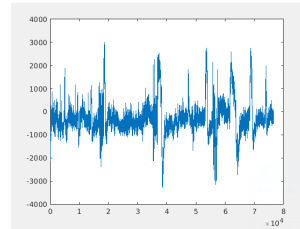


Fig. 2. Raw EEG Signal

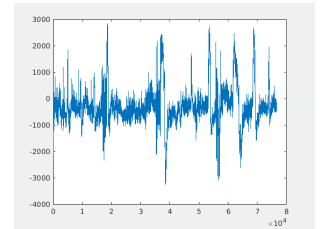


Fig. 3. EEG signal after wavelet denoising

LDA classifier is used as baseline model in the experimental study and its classification performance on the test set is the metric used to assess the different classification models. The experimental procedure consists of training and testing several classification algorithms on the BCI competition III dataset.

Six standard classifiers: LDA, Linear SVM, RBF-Kernel SVM, Artificial Neural-Network, Decision Tree and Logistic Regression are trained and have their respective hyper-parameters tuned using 5-fold cross validation. This cross validation scheme is adopted over 10-fold cross validation to reduce training time as the number of classifiers being used in the experiment are high. The classifiers are trained using a CSP feature extraction algorithm which results in a six-dimensional feature vector for each data point. Furthermore, the training dataset is comprised of 210 labeled instances that are split evenly between the two classes. The test dataset contains 34 594 data instances with 18 488 belonging to one class and 16 106 to the other. A six-dimensional feature vector is extracted for each test instance.

Since the number of data instances belonging to each class is different with roughly 2 000 more positive than negative examples, a second performance metric other than test error must be used since test error is a metric that is insensitive to randomness caused by different number of examples in each class. Hence Cohen's kappa value is also used for the experimental study of the selected classifiers. Using the test error and the Cohen's kappa value as classification metrics three performance comparisons are made: 1) baseline vs. other standard classifiers, 2) respective standard classifier (i.e. Decision Tree, Logistic Regression).

The results of the experimental results of processing the EEG data using the described processing pipeline with each of the five classifiers without utilization of wavelet signal denoising at the feature selection stage are summarized in Table 1 and the results with wavelet shrinkage are given in Table 2.

TABLE 1. CLASSIFICATION TEST RESULTS WITHOUT WAVELET

Classifier	Test Error %	Cohen's Kappa
LDA	13.02	0.737
Linear SVM	11.65	0.764
RBF-Kernel SVM	11.13	0.775
Decision Tree	16.53	0.671
Logistic Regression	12.66	0.745

TABLE 2. CLASSIFICATION TEST RESULTS WITH WAVELET

Classifier	Test Error %	Cohen's Kappa
LDA	12.18	0.754
Linear SVM	11.52	0.766
RBF-Kernel SVM	10.86	0.781
Decision Tree	15.65	0.687
Logistic Regression	12.55	0.748

The obtained test error and kappa value for the evaluated classifiers without and with wavelet denoising are shown on Fig. 4–7. Linear SVM and RBF-Kernel SVN classifiers show better performance results. The application of a linear maximum margin classifier (Linear SVM) improved the classification performance by roughly 1.5%.

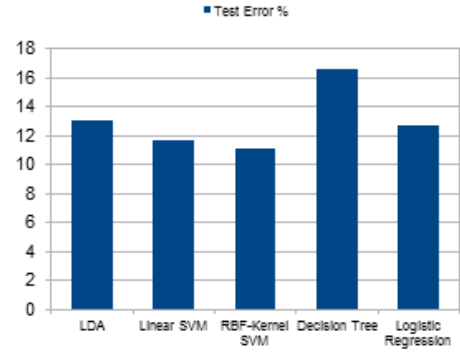


Fig. 4. Test Error without wavelet signal denoising

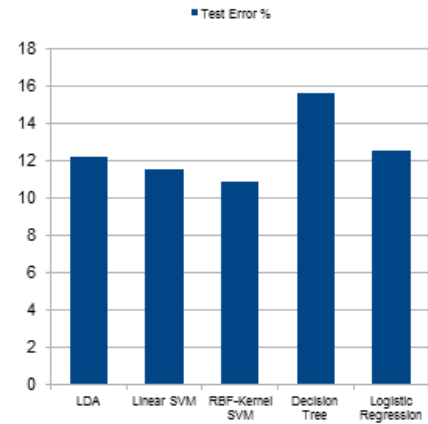


Fig. 5. Test Error with Wavelet Signal Denoising

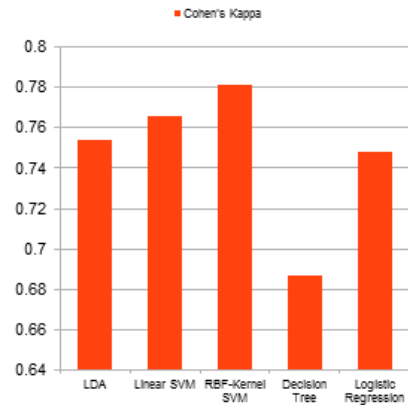


Fig. 6. Cohen's Kappa without Wavelet Signal Denoising

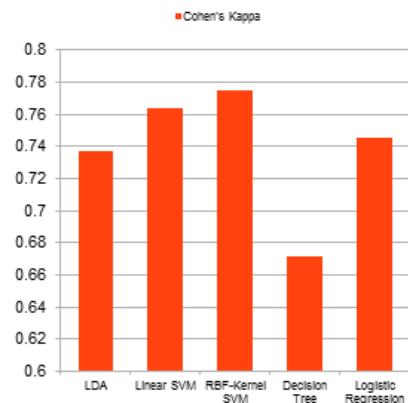


Fig. 7. Cohen's Kappa with Wavelet Signal Denoising

Also given that the training dataset has low feature dimension (6 dimensions) but a much greater amount of training instances the non-linear and more complex RBF-Kernel based SVM is the best performing standard classifier on the experimental dataset with a test error of 11.13% and kappa value of 0.775 without wavelet shrinkage and a test error of 10.86% and kappa value of 0.781 with wavelet based feature selection included in the processing pipeline thus giving and improved results for the examined dataset.

VI. CONCLUSION

In this work we examined the use of five classifiers LDA, Linear SVM, RBF-Kernel SVM, Decision Tree, and Logistic Regression for EEG based 2-class motor imagery classification problem utilizing a wavelet based signal denoising. The experimental results show improvement of the classification accuracy when wavelet shrinkage is applied at the feature selection stage of the data processing pipeline. Future work on the topic will be based on utilization of data denoising at the preprocessing stage of the data processing pipeline as well as more robust feature extraction techniques to deal with signal denoising as utilization of metaheuristics based algorithms.

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